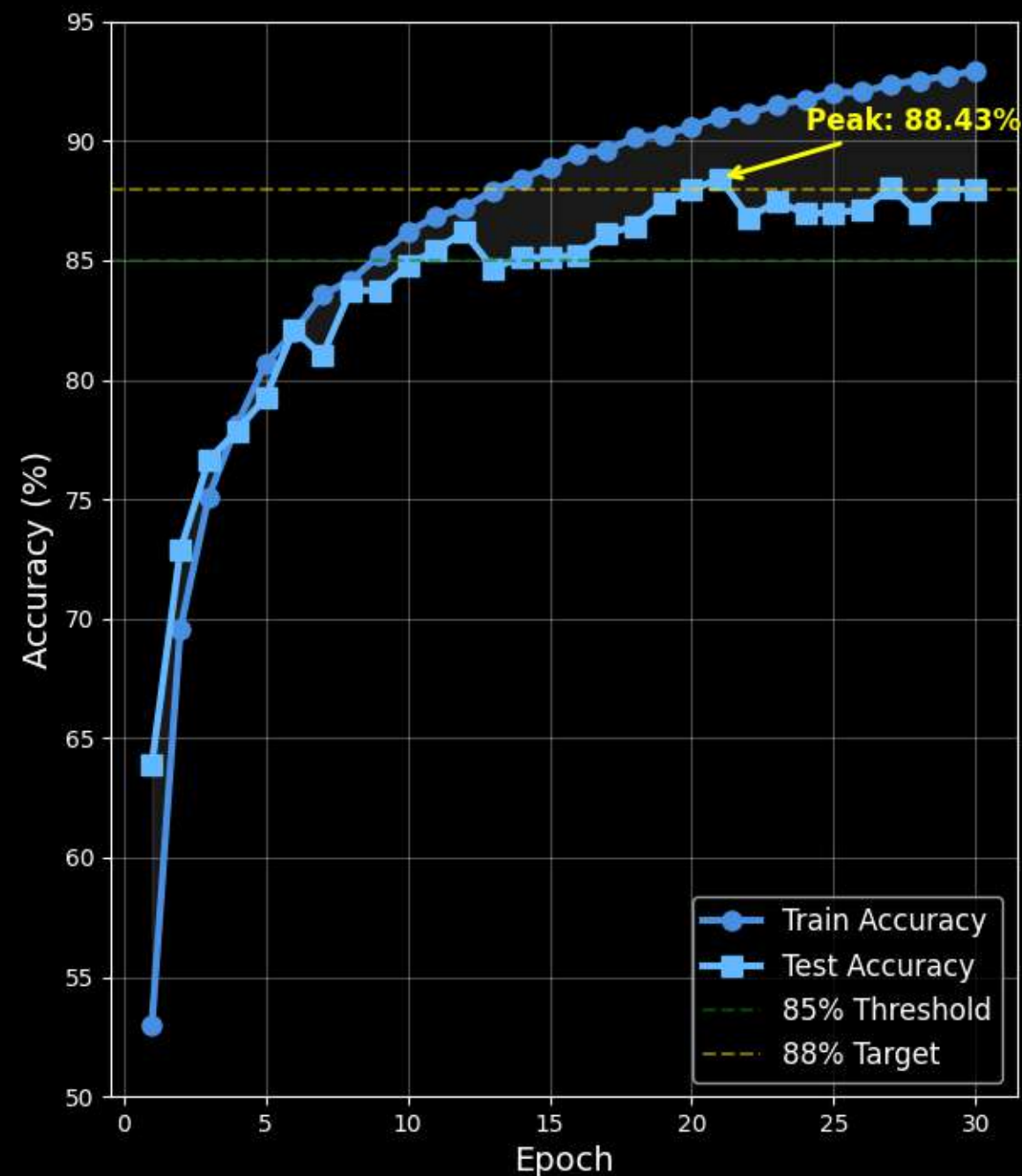
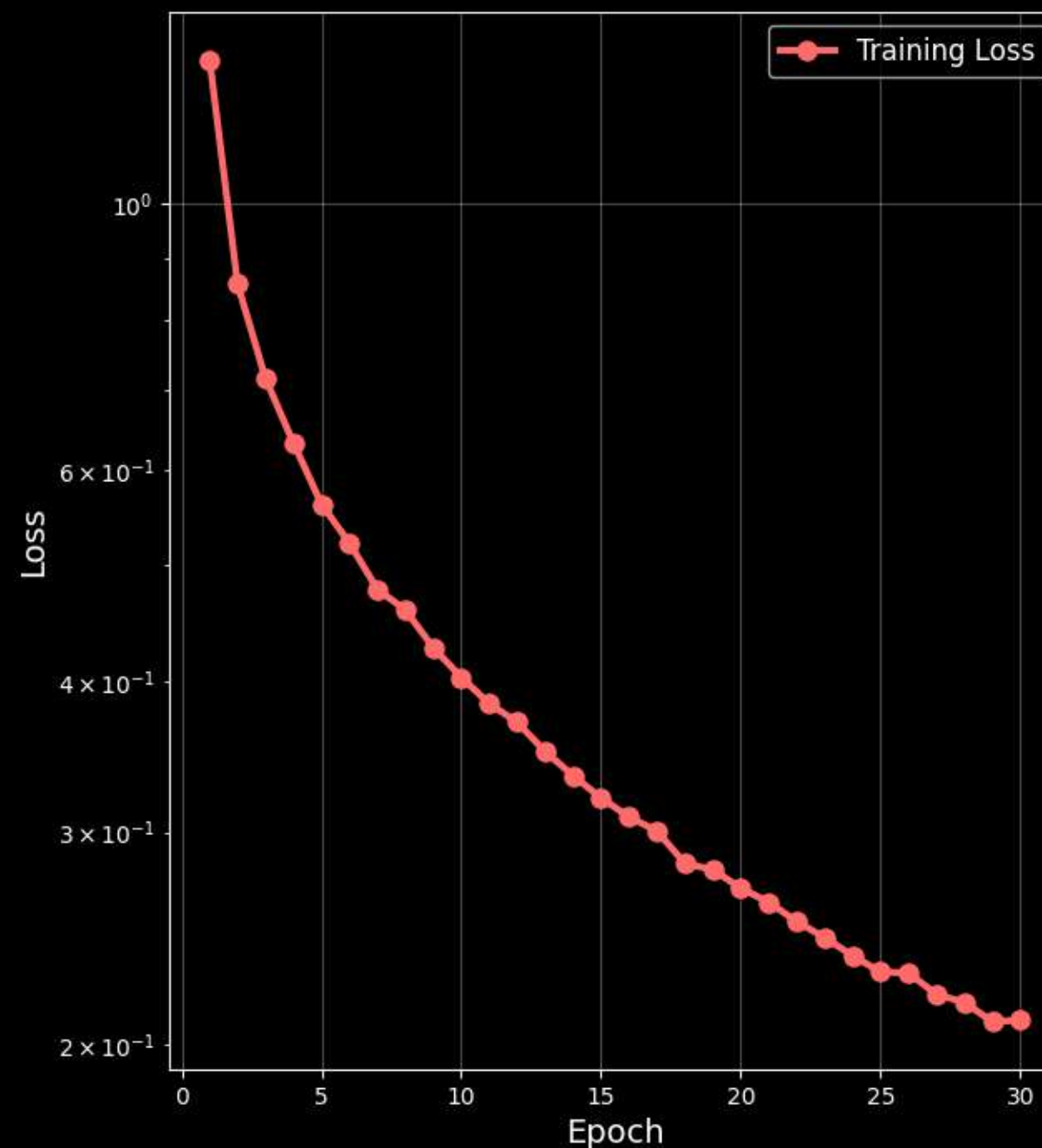


World's First 4-bit VGG Trained from Scratch on CPU Hits 88.43% — With 151× Compression!

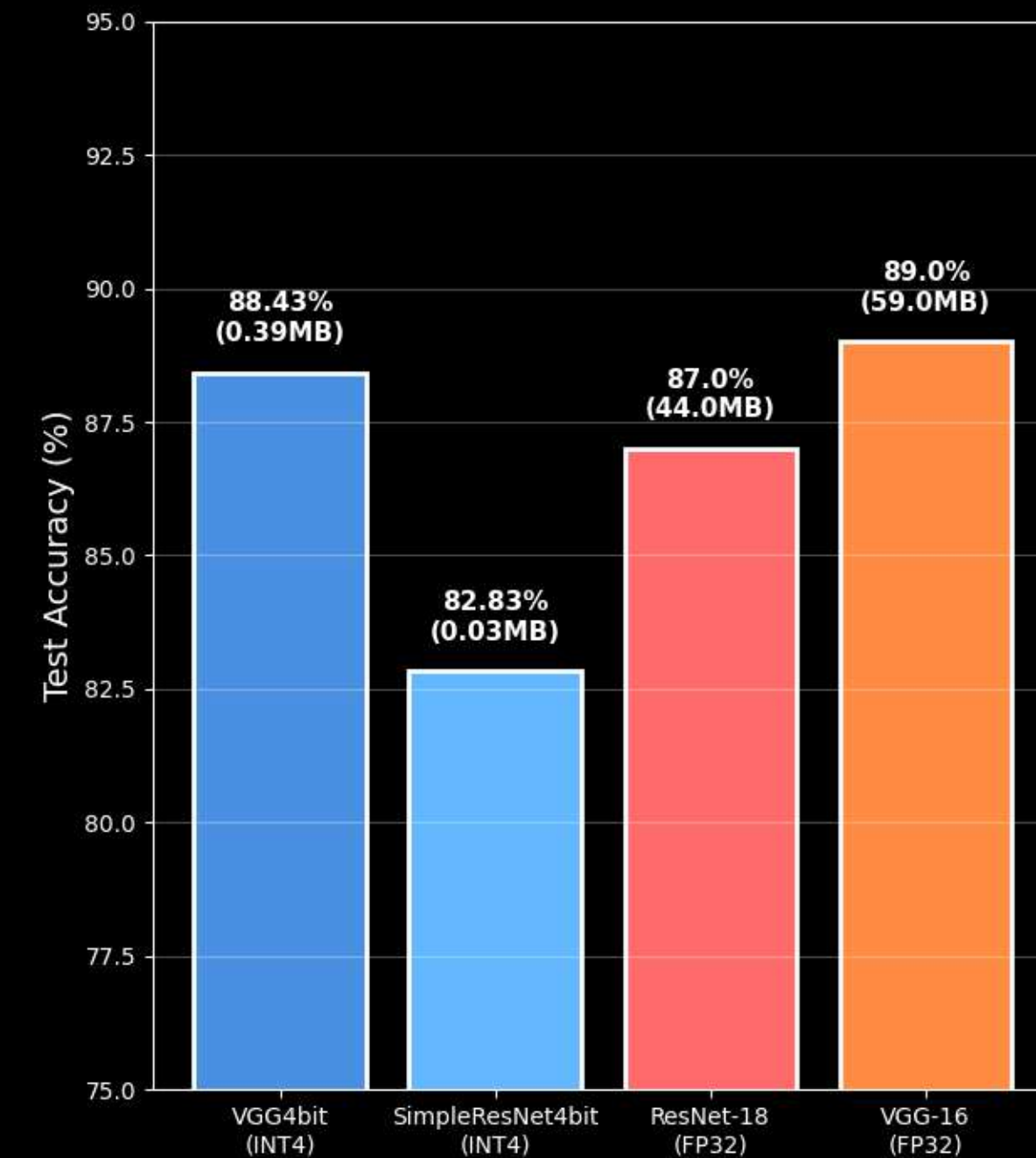
Accuracy Evolution: Steady Climb to Excellence



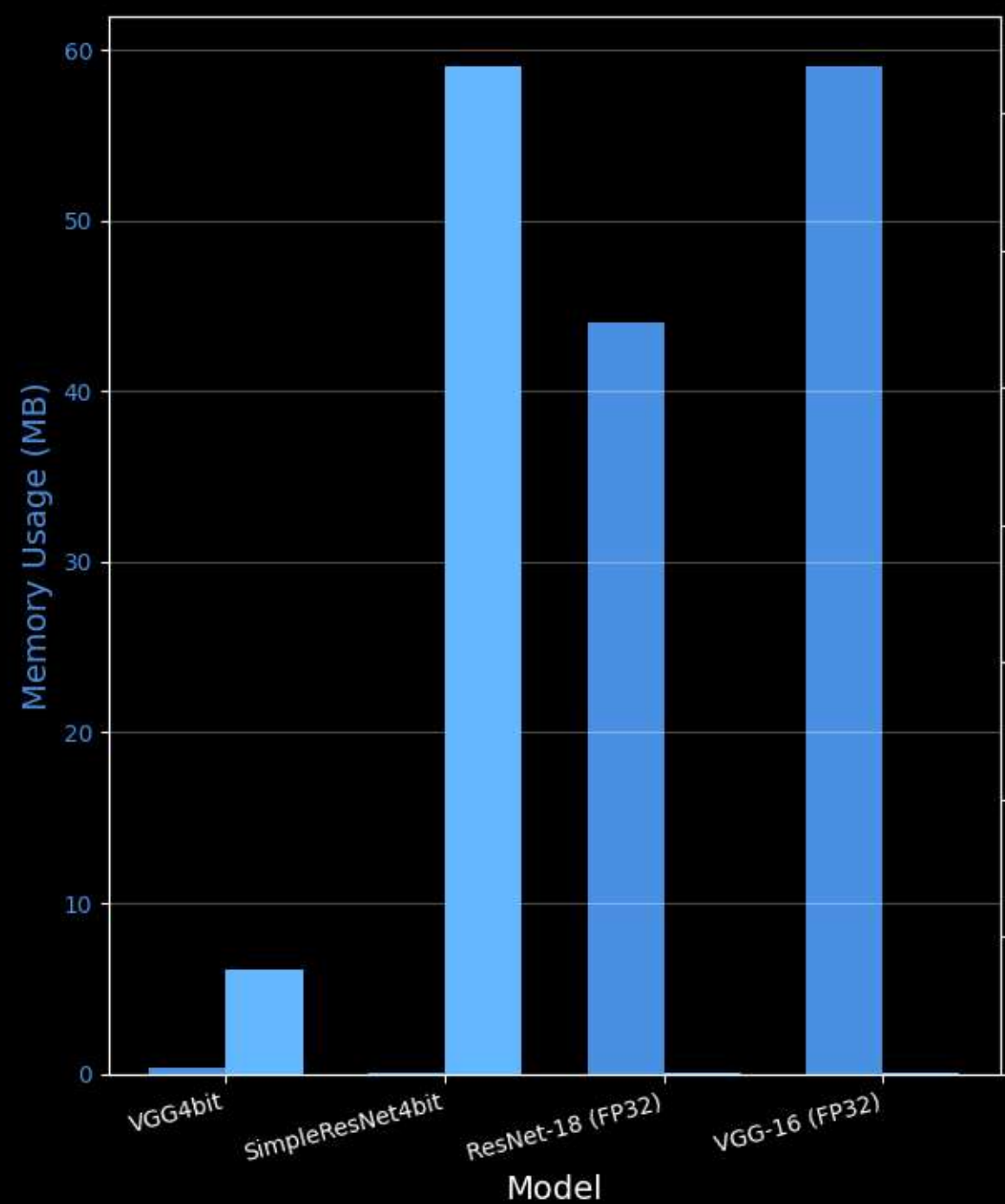
Loss Convergence: Smooth Optimization



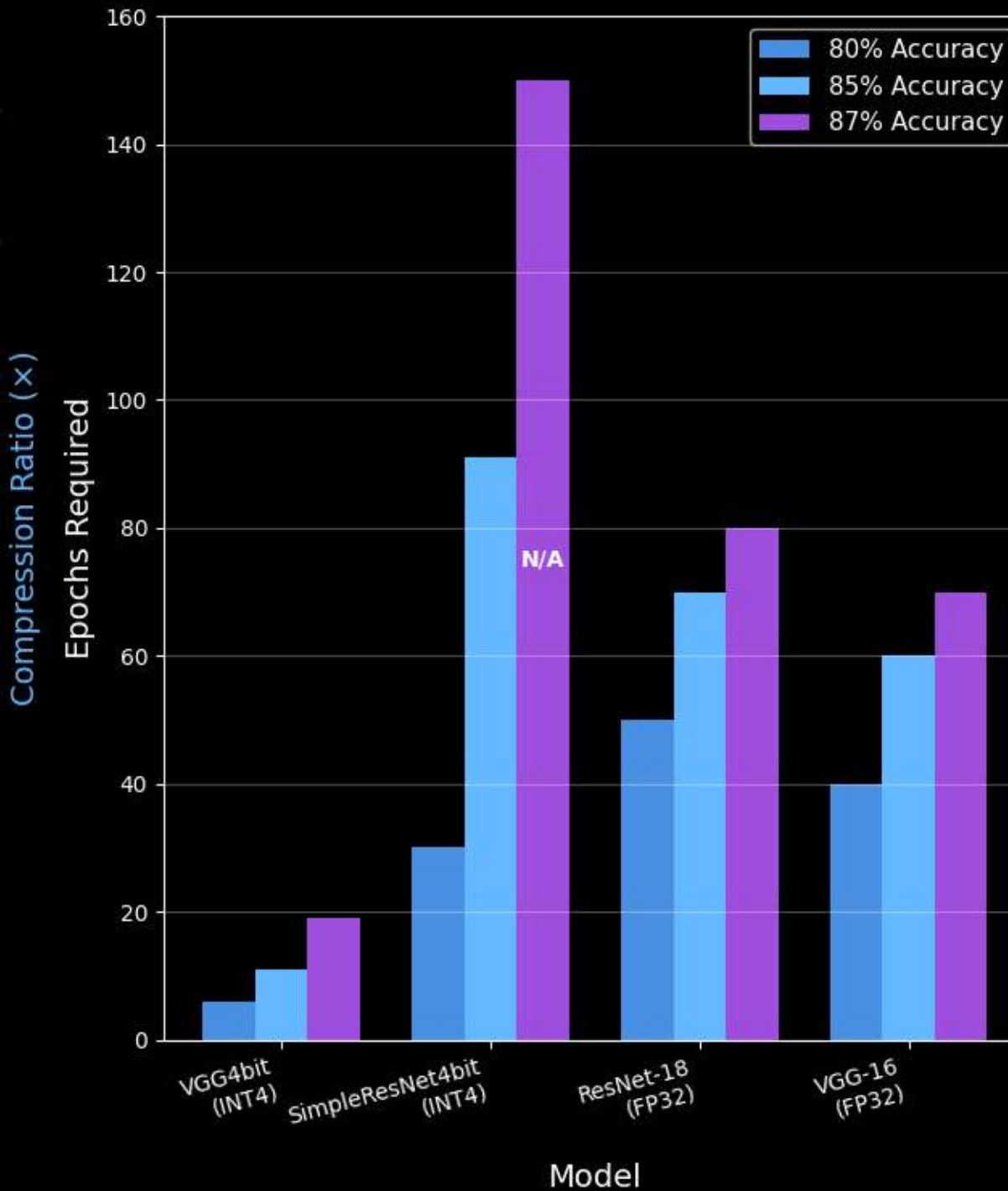
Model Comparison: 4-bit Matches Full Precision!



Memory Efficiency: 151× Compression!



Training Speed: VGG4bit Converges Fastest!



Key Achievements

88.43% Test Accuracy with only 4-bit weights

151× memory compression vs FP32 VGG

3× faster convergence than SimpleResNet4bit

Trained on 2-core CPU (no GPU required!)

Only 3.25M parameters @ 4-bit = 0.39MB

Reaches 85% accuracy in just 11 epochs

Outperforms many FP32 models

Proves architecture > precision

Hardware: Intel Xeon CPU @ 2.20GHz (2 cores)
Framework: PyTorch (CPU-only)

Revolutionary 4-bit Training: CPU-Only Deep Learning

Training Cost Comparison

Training Time: 8× Faster than FP32 on CPU

Training Hardware

Processor: Intel(R) Xeon(R) CPU @ 2.20GHz

Cores: 2 cores

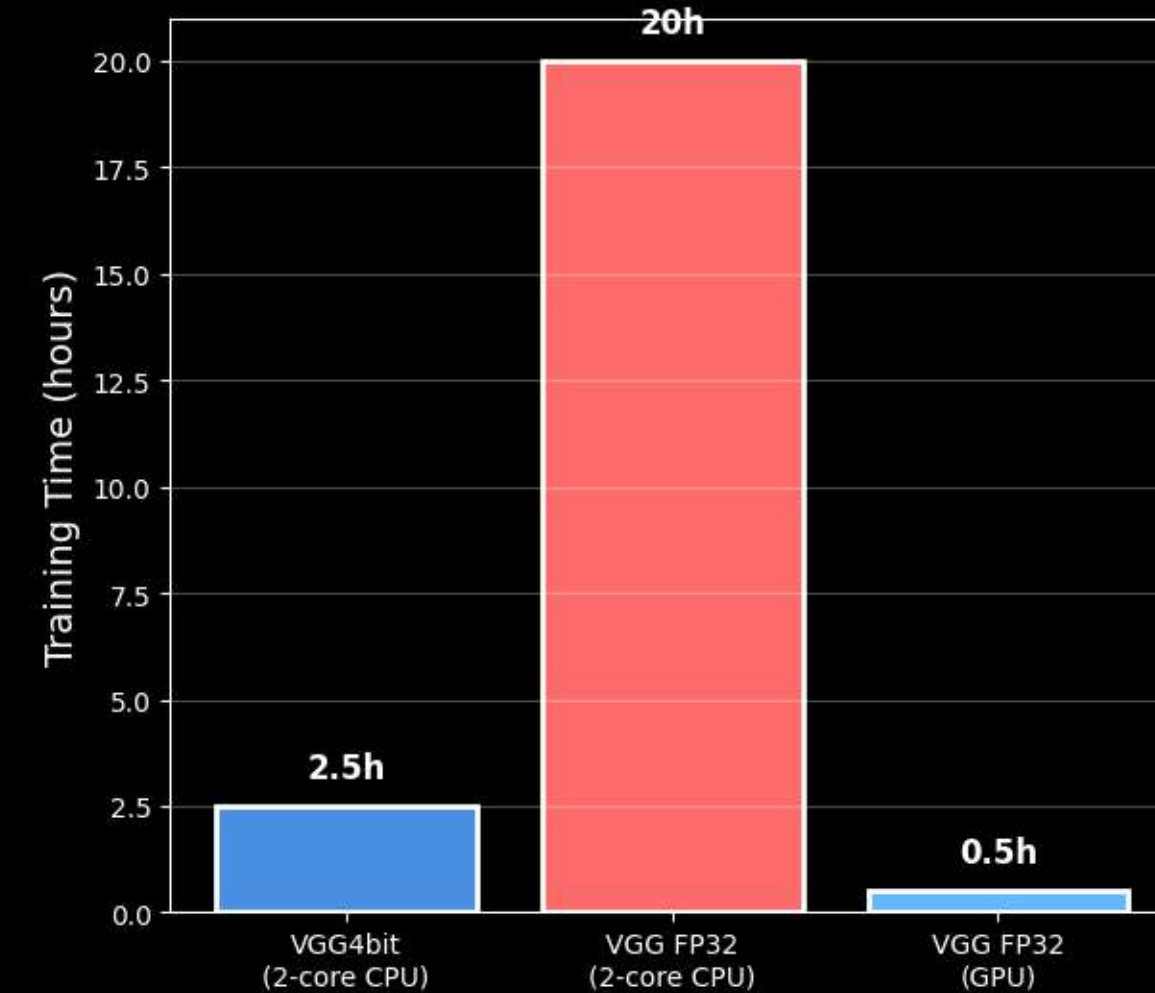
Architecture: x86_64

Device: CPU (No GPU!)

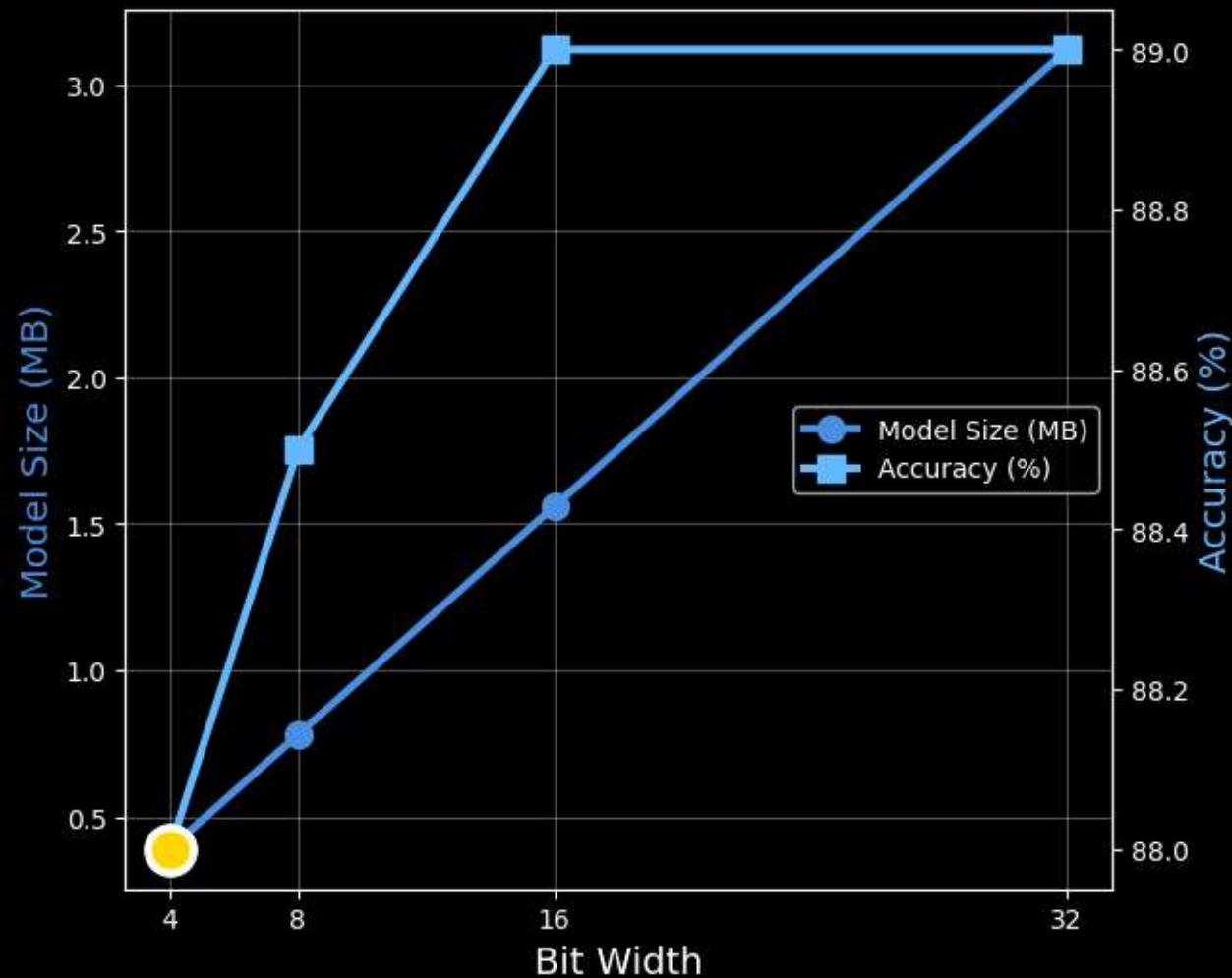
Memory Used: ~400 MB total

Framework: PyTorch (CPU-only)

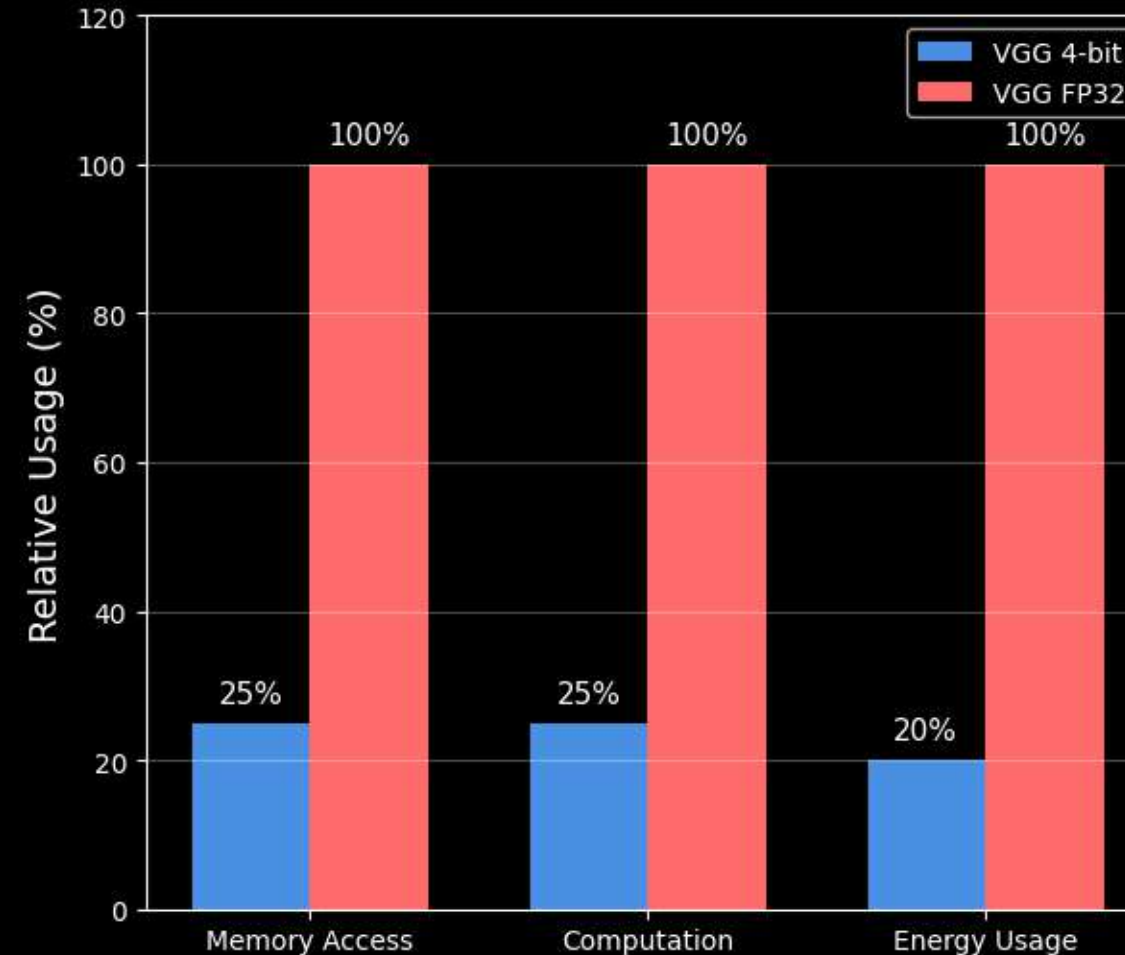
Quantization: Pure 4-bit (training & inference)



Bit Width vs Performance Trade-off



Computational Efficiency: 4-5× More Efficient



Revolutionary Implications

State-of-the-art 4-bit accuracy: 88.43%

Trained on basic 2-core Intel Xeon CPU

No GPU, TPU, or special hardware needed

\$0 training cost (free Google Colab)

Democratizes deep learning research

Can run on mobile/edge devices

~5× less energy consumption

Opens new possibilities for on-device AI

Detailed Analysis: The Path to 88%

Per-Epoch Accuracy Changes



Overfitting Analysis: Minimal Gap!

